

Software Requirements Specification

Project: Spotter

Version 1.2.0

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1. Introduction

1.1 Purpose

The purpose of this document is to build an online software system that helps drivers locate, reserve, and pay for parking spaces and parking garages in real time. The platform will be available through web and mobile applications and will provide parking availability information, navigation assistance, and management tools for parking operators.

The goal is to develop a project management plan for the creation of this system, including requirements gathering, stakeholder management, scheduling, budgeting, risk management, and team coordination. Drivers often spend significant time searching for available parking, leading to congestion, frustration, and lost productivity. Parking operators need better tools to monitor occupancy, manage pricing, and improve utilization of parking assets.

1.2 Scope

1.2.1 Overview and In Scope

The service is a cross-platform parking application (iOS, Android, and web) that unifies two supply models under a single search-and-booking experience:

(Layer 1) Commercial Parking Booking: a reservation and payment system for professionally operated or municipally regulated parking spaces.

- User registration, authentication, login, and vehicle profile management
- Real-time interactive map-based search and price comparison across parking types: garages, lots, event/venue parking, airport parking, and metered/on-street parking
- Advance reservation and digital payment processing
- Digital pass generation (QR/barcode) for facility entry

- Session management to view, extend, or cancel an active/upcoming reservation
- Push/email notifications and alerts for session start, expiration, and reminders
- Reservation history, receipts, and saved/favorite locations
- Administrative dashboard and occupancy reporting and analytics for parking operators

(Layer 2) Peer-to-Peer Driveway Marketplace: a supplementary marketplace allowing homeowners near social hubs (arenas, airports, downtown districts) to list private driveway space for driver booking.

- Mockups/wireframes for host listing creation, driveway search results integrated into the main map view, booking confirmation, and a basic two-sided rating display
- Written user flow documentation describing how a host lists a space and how a driver discovers and books it
- No functional backend, payment processing, host verification system, or liability/insurance handling will be implemented for this layer in the current release; these are documented as future-phase requirements

1.2.2 Out of Scope

- Full host onboarding, identity verification, and payout processing
- Insurance/liability management or dispute resolution workflows
- Business/fleet parking account management
- In-car system integrations (e.g., CarPlay, Android Auto)
- AI-based customer support or chat features
- Real-time IoT sensor integration for live space availability

2. Overall Description

2.1 User Classes and Characteristics

Drivers (Primary Users)

Volume: Largest user group, majority of total active users

Role: Search for, reserve, and pay for parking across commercial (garage/lot/metered) and P2P driveway listings, manage active sessions

Technical proficiency: Basic smartphone/web literacy, no training required

Frequency of use: High/daily to weekly, depending on commute/travel habits

Access level: Standard registered-user permissions

Homeowner Hosts (P2P Supply Providers)

Volume: Small relative to Drivers, concentrated near target social hubs (arenas, airports, downtown districts)

Role: List private driveway/parking space, set availability and pricing; view incoming bookings

Technical proficiency: Basic, onboarding designed to require minimal setup effort

Frequency of use: Low to moderate, periodic listing updates rather than daily engagement

Access level: Host-tier permissions (listing management), no access to Driver-side booking tools beyond their own listings

Guest Users (Unregistered)

Volume: Variable, represents pre-conversion traffic (browsing before signup)

Role: Search and compare pricing only; cannot complete a reservation or payment

Technical proficiency: Basic

Frequency of use: Typically one-time or infrequent, until conversion to registered Driver

Access level: Restricted/read-only, until sign up

Parking Operators

Volume: Small, typically one operator account per managed facility, garage, or lot

Role: Monitor real-time occupancy, view reservation and revenue analytics, and adjust pricing for their managed facility/facilities via the operator dashboard

Technical proficiency: Basic to moderate, comfortable with standard business/web dashboard tools, no specialized training required

Frequency of use: Daily to weekly, operational check-ins rather than continuous engagement

Access level: Facility-scoped operator permissions = access limited to data and settings for facilities they manage. No visibility into other operators' data or Driver/Host account information

System Administrators

Volume: Very small, internal staff only

Role: Manage user accounts, resolve support issues, moderate host listings, monitor system health

Technical proficiency: Moderate to advanced, expected to use internal admin tooling

Frequency of use: Ongoing/operational

Access level: Elevated/administrative permissions

2.2 Operating Environment

Client Platforms

- Mobile: iOS and Android, delivered as native or cross-platform apps

- Web: Modern browsers (Chrome, Safari, Firefox, Edge) on desktop and mobile web

Server/Backend

- Cloud-hosted backend (e.g., AWS/Azure) using a RESTful API architecture
- Relational (MySQL) database for user, listing, and transaction data
- Standard load-balanced, horizontally scalable deployment for variable traffic

Third-Party Integrations

- Mapping/geolocation service provider for search, discovery, and directions
- Payment gateway provider for processing card and mobile wallet transactions
- Push notification service for session/reminder alerts
- (future phase) Identity verification service for Layer 2 host onboarding

Network Dependency

- Requires an active internet connection for all core functions; no offline mode is in scope

2.3 Design and Implementation Constraints

1. Fixed semester-long timeline
2. Dependence on third-party services and APIs
3. Compliance with data security and privacy requirements (PCI-DSS)
4. Platform pairing and consistent functionality across iOS/Android and web
5. Active network dependency
6. Variable volume scalability
7. Layer 2 features are constrained to mockup/prototype fidelity.

2.4 Assumptions and Dependencies

Assumptions

- Unlimited theoretical budget
- Users have access to a smartphone/web-enabled device and an active internet connection.
- Drivers and Hosts have a valid payment method and legal authority to list their property.
- Third-party mapping and payment services remain available, stable, and within their published API terms throughout development and testing.
- Layer 2 (P2P) functionality is assumed to remain at mockup/prototype for the duration of this project and will not require live transaction support.
- Local zoning/HOA restrictions on residential parking rentals are assumed to be a future-phase concern, not something the current system needs to enforce or validate.

Dependencies

- System functionality depends on the continued availability of the chosen mapping/geolocation service provider.
- Payment processing depends on a third-party payment gateway and its compliance infrastructure.
- Push notifications depend on platform-level notification services (e.g., APNs, FCM) for mobile, and browser notification APIs for web.
- Cross-platform consistency depends on a shared backend/API layer being implemented before mobile and web clients are built out.
- Any future expansion of Layer 2 to full functionality depends on adding identity verification and payout processing services not currently in scope.

3. Functional Requirements

FR1: The system shall allow users to register, log in, and manage a vehicle profile.

FR2: The system shall allow users to search for parking by location and time, displaying results on a map view.

FR3: The system shall display available parking types with associated pricing.

FR4: The system shall allow users to compare prices across nearby parking options before booking.

FR5: The system shall allow users to reserve and prepay for a parking session via credit/debit card or mobile wallet.

FR6: The system shall generate a digital pass (QR/barcode) upon successful booking for entry.

FR7: The system shall allow users to view, extend, or cancel an active or upcoming reservation.

FR8: The system shall send notifications for session start, upcoming expiration, and reminders.

FR9: The system shall maintain a booking history/receipts and allow users to save favorite locations.

FR10: The system shall display prototype-level P2P driveway listings (mockup data) integrated into the main search/map view.

FR11: The system shall provide a mockup host listing creation flow (non-functional data entry, no live backend persistence required).

FR12: The system shall display a mockup two-sided rating/review interface for Layer 2 bookings.

FR13: The system shall provide parking operators with a dashboard displaying real-time occupancy for their managed facility/facilities.

FR14: The system shall allow parking operators to view reservation and revenue history for their facility over a selectable date range.

FR15: The system shall allow parking operators to view and adjust pricing for their listed parking spaces, subject to system-defined limits.

FR16: The system shall generate basic analytics (e.g., average occupancy rate, peak usage hours, total reservations) for a parking operator's facility.

FR17: The system shall restrict dashboard access to authenticated parking operator accounts, limited to data for facilities they manage.

4. Non-Functional Requirements

NFR1: The system shall be available on iOS/Android and web platforms with consistent core functionality.

NFR2: The system shall respond to search queries within 2 seconds under normal load conditions.

NFR3: The system shall scale horizontally to handle demand spikes during high-traffic events.

NFR4: The system shall encrypt all payment and personal data in transit and at rest, complying with PCI-DSS standards.

NFR5: The system shall maintain 99% uptime for core booking functionality during business/operational hours.

NFR6: The system's UI shall be usable by individuals with basic smartphone/web literacy.

NFR7: The system shall log and handle failures from internal mapping and payment service abstractions gracefully, without exposing raw third-party errors to end users.

NFR8: The system shall support at least the two most recent major OS versions for iOS/Android, and current web browsers.

NFR9: The system shall not require offline functionality, but shall fail with a clear error statement when no network connection is present.

NFR10: The system's architecture shall allow Layer 2 to be extended from prototype to full functionality without requiring a redesign of the shared backend/data model.

NFR11: The operator dashboard shall reflect occupancy data with no more than a 1-minute delay from the underlying reservation system.

NFR12: The operator dashboard shall be accessible via web only, consistent with an internal/business-tool use case rather than a consumer-facing feature.

NFR13: Analytics data displayed on the dashboard shall be scoped and access-controlled so that one operator cannot view another operator's facility data.

5. Use Case Examples

Title: Register Account

Actor: Driver, Host

Precondition: User has downloaded the app or accessed the web platform and is not yet registered

Steps:

1. User selects "Sign Up"
2. User enters name, email, and password
3. System validates input and creates account
4. System sends confirmation notification

Postcondition: User account is created and user is logged in

Title: Log In

Actor: Driver, Host, Administrator

Precondition: User has a registered account

Steps:

1. User enters email and password
2. System validates credentials
3. System grants access to the appropriate dashboard based on user role

Postcondition: User is authenticated and directed to their home screen

Title: Guest Browses Parking Availability

Actor: Guest User

Precondition: User has not logged in or is registered

Steps:

1. Guest opens the app/web platform
2. Guest enters a location, date, and timeframe
3. System displays available parking options and pricing on a map
4. Guest attempts to book and is prompted to register/log in

Postcondition: Guest views available parking but cannot complete a reservation

Title: Manage Vehicle Profile

Actor: Driver

Precondition: Driver is logged in

Steps:

1. Driver navigates to profile settings
2. Driver adds or edits vehicle details (license plate, vehicle type)
3. System saves updated profile

Postcondition: Vehicle profile is updated and available for future bookings

Title: Search for Parking

Actor: Driver

Precondition: Driver is logged in

Steps:

1. Driver enters a destination or enables location services

2. System queries available parking spaces (garages, lots, metered, event/airport)
3. System displays results on a map view

Postcondition: Driver views a list/map of nearby available parking options

Title: Compare Parking Prices

Actor: Driver

Precondition: Search results have been returned

Steps:

1. Driver views multiple parking options in the results list
2. System displays address, price, distance, walking time, and duration for each option
3. Driver selects a preferred option

Postcondition: Driver has identified a chosen parking option to book

Title: Reserve and Pay for Parking

Actor: Driver

Precondition: Driver has selected a parking option and is logged in

Steps:

1. Driver confirms reservation start/end time
2. Driver confirms vehicle profile and payment method
3. System processes payment through the payment service
4. System confirms the reservation

Postcondition: Reservation is created and payment is processed

Title: View Digital Pass for Entry

Actor: Driver

Precondition: Driver has an active reservation

Steps:

1. Driver opens the reservation from their booking list
2. System displays a QR code/barcode digital pass
3. Driver presents the pass at the facility entrance

Postcondition: Driver gains entry to the reserved parking space

Title: Extend Active Parking Session

Actor: Driver

Precondition: Driver has an active, ongoing reservation

Steps:

1. Driver selects "Extend Session" from the active reservation screen.
2. Driver selects additional time interval (e.g., add 1hr for \$1.50, add 2hrs for \$2.00)
3. System checks availability and processes additional payment
4. System updates the reservation end time

Postcondition: Reservation duration is extended

Cancel Reservation

Actor: Driver

Precondition: Driver has an upcoming or active reservation

Steps:

1. Driver selects "Cancel" on the reservation
2. System displays cancellation terms (e.g., refund eligibility)
3. Driver confirms cancellation
4. System processes refund if applicable and cancels the reservation

Postcondition: Reservation is canceled and reflected in booking history

Title: Receive Session Notification

Actor: Driver

Precondition: Driver has an active or upcoming reservation

Steps:

1. System monitors reservation status and time remaining
2. System triggers a notification at user-defined intervals (e.g, 10 minutes before reservation time)
3. Driver receives push/email notification

Postcondition: Driver is informed of session status in a timely manner

Title: View Booking History/Receipts and Manage Favorites

Actor: Driver

Precondition: Driver is logged in and has at least one past reservation

Steps:

1. Driver navigates to "Booking History"
2. System displays past and current reservations with payment history
3. Driver selects a location and marks it as a favorite

Postcondition: Location is saved for faster future booking

Title: Browse P2P Driveway Listings (Prototype)

Actor: Driver

Precondition: Driver is logged in and searching in an area with prototype Layer 2 listings

Steps:

1. Driver's search results include driveway listings alongside commercial options
2. Driver selects a driveway listing to view mockup details (photos, price, availability)
3. Driver views a mockup booking confirmation screen (non-functional)

Postcondition: Driver views prototype P2P booking flow; no live transaction occurs

Title: Create Host Listing (Prototype)

Actor: Host

Precondition: Host is logged in

Steps:

1. Host selects "List My Lot"
2. Host enters mockup listing details (address, photos, price, availability window, etc.)
3. System displays a confirmation screen showing the listing preview

Postcondition: A prototype listing preview is generated, no live persistence or public listing occurs

Title: Administrator Moderates Host Listing / Resolves Dispute

Actor: Administrator

Precondition: Administrator is logged in with elevated permissions

Steps:

1. Administrator reviews flagged host listings or user disputes
2. Administrator reviews relevant booking/account data
3. Administrator approves, removes, or escalates the listing/dispute

Postcondition: Listing status is updated or dispute is resolved/escalated

Title: View Facility Occupancy Dashboard

Actor: Parking Operator

Precondition: Operator is logged in and manages at least one facility in the system

Steps:

1. Operator logs in and navigates to the Dashboard
2. System retrieves current occupancy data for the operator's facility/facilities
3. System displays occupancy as a percentage or count (spaces filled/available)

Postcondition: Operator views real-time occupancy status for their facility

Title: View Reservation and Revenue Analytics

Actor: Parking Operator

Precondition: Operator is logged in and has historical reservation data

Steps:

1. Operator navigates to the Analytics section of the dashboard
2. Operator selects a date range
3. System displays reservation counts, revenue totals, and peak usage hours for that range

Postcondition: Operator views summarized analytics for the selected period.

Title: Adjust Facility Pricing

Actor: Parking Operator

Precondition: Operator is logged in and manages a facility with existing pricing data

Steps:

1. Operator navigates to facility settings
2. Operator updates pricing for one or more parking types (hourly, daily, event)
3. System validates the input against defined pricing limits
4. System saves and applies the updated pricing to future searches/bookings

Postcondition: Updated pricing is reflected in the system for new reservations

6. Work Breakdown Structure (App Functionalities)

1. Frontend

1.1 Authentication UI

1.1.1 Sign-up form

1.1.2 Login form

1.1.3 Password reset

1.1.4 Role-based dashboard redirect (Driver/Host/Operator/Admin)

1.2 Home/Map View

1.2.1 Interactive map render

1.2.2 Location permission/geolocation prompt

1.2.3 Navigation bar and menu

1.2.4 Guest vs logged-in state handling

1.3 Parking Search

1.3.1 Destination input (autocomplete)

1.3.2 Date/time selector

1.3.3 Results list view

1.3.4 Map pin rendering (commercial + P2P listing types)

1.3.5 Filters (price, distance, parking type)

1.3.6 Listing detail view (address, price, walking time, photos)

1.4 Booking Flow

1.4.1 Reservation day/time confirmation

1.4.2 Vehicle profile selection

1.4.3 Payment method selection

- 1.4.4 Booking confirmation screen
- 1.5 Reservation Management
 - 1.5.1 Active/upcoming reservation list
 - 1.5.2 Digital pass display (QR/barcode)
 - 1.5.3 Extend session UI
 - 1.5.4 Cancel reservation UI with terms display
- 1.6 User Profile
 - 1.6.1 Account settings
 - 1.6.2 Vehicle profile add/edit
 - 1.6.3 Saved payment methods
 - 1.6.4 Booking history and receipts
 - 1.6.5 Favorite locations
- 1.7 Notifications UI
 - 1.7.1 In-app notification display
 - 1.7.2 Notification preference settings
- 1.8 Operator Dashboard (web only)
 - 1.8.1 Occupancy display
 - 1.8.2 Analytics view (date range selector, charts)
 - 1.8.3 Pricing adjustment form
 - 1.8.4 Facility settings
- 1.9 Admin Console
 - 1.9.1 User account management view
 - 1.9.2 Flagged listing, dispute queue

1.9.3 System health monitoring view

1.10 P2P Layer (Prototype/Mockup)

1.10.1 Host listing creation screens

1.10.2 Driveway listing detail mockup

1.10.3 Mockup booking confirmation

1.10.4 Two-sided rating/review display

2.0 Backend

2.1 Authentication & Authorization

2.1.1 User registration service

2.1.2 Login/session token handling

2.1.3 Role-based access control (Driver/Host/Operator/Admin)

2.1.4 Facility-scoped operator permissions

2.2 User Services

2.2.1 Profile CRUD

2.2.2 Vehicle profile CRUD

2.2.3 Favorites management

2.3 Parking Inventory Service

2.3.1 Facility/space data model

2.3.2 Availability lookup

2.3.3 Pricing rules engine

2.3.4 Parking type handling (garage, lot, metered, event, airport)

2.4 Search Service

2.4.1 Location-based query handling

2.4.2 Results ranking/sorting

2.4.3 Price comparison logic

2.5 Reservation Service

2.5.1 Create reservation

2.5.2 Extend reservation

2.5.3 Cancel reservation and refund eligibility

2.5.4 Reservation state management

2.5.5 Digital pass generation (QR/barcode)

2.6 Payment Service

2.6.1 Payment gateway integration layer

2.6.2 Transaction processing

2.6.3 Refund processing

2.6.4 Receipt generation

2.7 Notification Service

2.7.1 Scheduled session reminders

2.7.2 Push dispatch (APNs/FCM)

2.7.3 Email dispatch

2.8 Analytics Service

2.8.1 Occupancy aggregation

2.8.2 Reservation/revenue reporting

2.8.3 Peak usage calculation

2.9 Admin Services

2.9.1 Account moderation actions

2.9.2 Listing moderation

2.9.3 Dispute record handling

2.10 Error & Logging

2.10.1 Third-party failure handling/fallbacks

2.10.2 System logging

2.10.3 Offline/no-network error states

3.0 Data Layer

3.1 User & account tables

3.2 Vehicle profile tables

3.3 Facility/parking space tables

3.4 Reservation tables

3.5 Transaction/payment records

3.6 Analytics/occupancy records

3.7 P2P listing schema (future phase)

4.0 Integrations

4.1 Mapping/geolocation service

4.2 Payment gateway

4.3 Push notification service

4.4 Identity verification (future phase)

7. Timeline Draft

 Spotter Gantt Chart - Sheet1.pdf

8. Kanban Board for Sprint 1 (Jira)

Spaces

Spotter

...

Summary

Backlog

Board

Development

Timeline

Docs

Forms

+

Search backlog...

IC

Filter

Epic

No epic

> Login

> UI: Op/User

> Backend Process

> Reporting

+ Create epic

SCRUM Sprint 1 24 Sep – 8 Oct (45 work items)

000 Complete sprint

SPT-9 User Registration (Driver/Host si... Login To Do

SPT-10 User Login (credential validation) Login To Do

SPT-11 Password Reset via Email Link Login To Do

SPT-12 Role-Based Dashboard Redirect Login To Do

SPT-13 Role-Based Dashboard Redirect Login To Do

SPT-14 Operator Dashboard Layout/Shell UI: Op/Us... To Do

SPT-15 Occupancy Display Component UI: Op/Us... To Do

SPT-16 Analytics View (charts, date ran... UI: Op/Us... To Do

SPT-17 Pricing Adjustment Form UI: Op/Us... To Do

SPT-18 Facility Settings Page UI: Op/Us... To Do

SPT-19 Operator Navigation Menu UI: Op/Us... To Do

SPT-20 Operator Dashboard Web Layout UI: Op/Us... To Do

SPT-21 Home / Map View UI UI: Op/Us... To Do

SPT-22 Parking Search Bar (autocompl... UI: Op/Us... To Do

SPT-23 Search Results List & Map Pins UI: Op/Us... To Do

SPT-24 Booking Flow Screens UI: Op/Us... To Do

SPT-25 Reservation Management (list, ... UI: Op/Us... To Do

SPT-26 Digital Pass Display (QR/barcod... UI: Op/Us... To Do

SPT-27 User Profile & Vehicle Profile UI UI: Op/Us... To Do

SPT-28 Host Listing Creation UI (Protot... UI: Op/Us... To Do

27

☒	SPT-31	Occupancy Aggregation Service	Backend ...	To Do	-	=	IC
☒	SPT-32	Analytics Reporting Service	Backend ...	To Do	-	=	IC
☒	SPT-33	Pricing Rules Engine	Backend ...	To Do	-	=	IC
☒	SPT-34	Facility-Scoped Access Control	Backend ...	To Do	-	=	IC
☒	SPT-35	Facility/Space Data Model	Backend ...	To Do	-	=	IC
☒	SPT-36	Operator Auth/Session Backend	Backend ...	To Do	-	=	IC
☒	SPT-37	Peak Usage Calculation Logic	Backend ...	To Do	-	=	IC
☒	SPT-38	Reservation Service (create/ext...	Backend ...	To Do	-	=	IC
☒	SPT-39	Payment Gateway Integration	Backend ...	To Do	-	=	IC
☒	SPT-40	Digital Pass Generation Service	Backend ...	To Do	-	=	IC
☒	SPT-41	Search Service Backend	Backend ...	To Do	-	=	IC
☒	SPT-42	Notification Service Backend	Backend ...	To Do	-	=	IC
☒	SPT-43	Vehicle Profile Backend (CRUD)	Backend ...	To Do	-	=	IC
☒	SPT-44	Favorites/Booking History Back...	Backend ...	To Do	-	=	IC
☒	SPT-45	Host Listing Mockup Backend S...	Backend ...	To Do	-	=	IC
☒	SPT-46	Occupancy Report Generation	Reporting	To Do	-	=	IC
☒	SPT-47	Revenue Report Generation	Reporting	To Do	-	=	IC
☒	SPT-48	Reservation Count Report	Reporting	To Do	-	=	IC
☒	SPT-49	Peak Hours Report	Reporting	To Do	-	=	IC
☒	SPT-50	Exportable Report Format (CSV...	Reporting	To Do	-	=	IC
☒	SPT-51	Booking History Report	Reporting	To Do	-	=	IC
☒	SPT-52	Payment/Receipt Report	Reporting	To Do	-	=	IC
☒	SPT-53	Favorite Locations Summary	Reporting	To Do	-	=	IC
☒	SPT-54	Session Notification Log	Reporting	To Do	-	=	IC
☒	SPT-55	Cancellation/Refund Report	Reporting	To Do	-	=	IC
+ Create							